

NTA JEE Mains Jan 2026

Test Date	24/01/2026
Test Time	9:00 AM - 12:00 PM
Subject	B. Tech

Section : Mathematics Section A

Q.1

If $\cot x = \frac{5}{12}$ for some $x \in \left(\pi, \frac{3\pi}{2}\right)$, then

$\sin 7x \left(\cos \frac{13x}{2} + \sin \frac{13x}{2} \right) + \cos 7x \left(\cos \frac{13x}{2} - \sin \frac{13x}{2} \right)$ is equal to

Options

1. $\frac{1}{\sqrt{13}}$
2. $\frac{5}{\sqrt{13}}$
3. $\frac{6}{\sqrt{26}}$
4. $\frac{4}{\sqrt{26}}$

Question Type : MCQ

Question ID : 444792537

Option 1 ID : 4447921833

Option 2 ID : 4447921832

Option 3 ID : 4447921830

Option 4 ID : 4447921831

Q.2

Let A(1, 0), B(2, -1) and C $\left(\frac{7}{3}, \frac{4}{3}\right)$ be three points. If the equation of the bisector

of the angle ABC is $\alpha x + \beta y = 5$, then the value of $\alpha^2 + \beta^2$ is

Options

1. 13
2. 8
3. 10
4. 5

Question Type : MCQ

Question ID : 444792538

Option 1 ID : 4447921837

Option 2 ID : 4447921835

Option 3 ID : 4447921836

Option 4 ID : 4447921834

Q.3

Consider an A.P.: $a_1, a_2, \dots, a_n$; $a_1 > 0$. If $a_2 - a_1 = \frac{-3}{4}$, $a_n = \frac{1}{4}a_1$, and

$$\sum_{i=1}^n a_i = \frac{525}{2}, \text{ then } \sum_{i=1}^{17} a_i \text{ is equal to}$$

Options 1. 476

2. 136

3. 238

4. 952

Question Type : MCQ

Question ID : 444792531

Option 1 ID : 4447921808

Option 2 ID : 4447921806

Option 3 ID : 4447921807

Option 4 ID : 4447921809

Q.4

Let $\alpha, \beta \in \mathbb{R}$ be such that the function $f(x) = \begin{cases} 2\alpha(x^2 - 2) + 2\beta x & , x < 1 \\ (\alpha + 3)x + (\alpha - \beta) & , x \geq 1 \end{cases}$

be differentiable at all $x \in \mathbb{R}$. Then $34(\alpha + \beta)$ is equal to

Options 1. 36

2. 48

3. 84

4. 24

Question Type : MCQ

Question ID : 444792543

Option 1 ID : 4447921855

Option 2 ID : 4447921856

Option 3 ID : 4447921857

Option 4 ID : 4447921854

Q.5

Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \vec{a} \times \vec{b}$. Let $\vec{d}$ be a vector such that

$|\vec{d} - \vec{a}| = \sqrt{11}$, $|\vec{c} \times \vec{d}| = 3$ and the angle between $\vec{c}$ and $\vec{d}$ is $\frac{\pi}{4}$. Then $\vec{a} \cdot \vec{d}$ is equal to

Options 1. 11

2. 0

3. 3

4. 1

Question Type : MCQ

Question ID : 444792540

Option 1 ID : 4447921845

Option 2 ID : 4447921842

Option 3 ID : 4447921844

Option 4 ID : 4447921843

Q.6 Let R be a relation defined on the set $\{1, 2, 3, 4\} \times \{1, 2, 3, 4\}$ by

$$R = \{((a, b), (c, d)) : 2a + 3b = 3c + 4d\}.$$

Then the number of elements in R is

Options 1. 15

2. 6

3. 12

4. 18

Question Type : MCQ

Question ID : 444792526

Option 1 ID : 4447921788

Option 2 ID : 4447921786

Option 3 ID : 4447921787

Option 4 ID : 4447921789

Q.7

If the function $f(x) = \frac{e^x (e^{\tan x - x} - 1) + \log_e (\sec x + \tan x) - x}{\tan x - x}$ is

continuous at $x = 0$, then the value of $f(0)$ is equal to

Options 1. $\frac{2}{3}$

2. $\frac{1}{2}$

3. 2

4. $\frac{3}{2}$

Question Type : MCQ

Question ID : 444792542

Option 1 ID : 4447921851

Option 2 ID : 4447921850

Option 3 ID : 4447921853

Option 4 ID : 4447921852

Q.8 Let a circle of radius 4 pass through the origin O, the points $A(-\sqrt{3}a, 0)$ and $B(0, -\sqrt{2}b)$, where a and b are real parameters and $ab \neq 0$. Then the locus of the centroid of $\triangle OAB$ is a circle of radius

- Options
1. $\frac{11}{3}$
 2. $\frac{7}{3}$
 3. $\frac{8}{3}$
 4. $\frac{5}{3}$

Question Type : **MCQ**

Question ID : **444792536**

Option 1 ID : **4447921829**

Option 2 ID : **4447921827**

Option 3 ID : **4447921828**

Option 4 ID : **4447921826**

Q.9 Let each of the two ellipses $E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, (a > b)$ and

$E_2 : \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1, (A < B)$ have eccentricity $\frac{4}{5}$. Let the lengths of the latus

recta of E_1 and E_2 be l_1 and l_2 , respectively, such that $2l_1^2 = 9l_2$. If the distance between the foci of E_1 is 8, then the distance between the foci of E_2 is

- Options
1. $\frac{8}{5}$
 2. $\frac{32}{5}$
 3. $\frac{96}{5}$
 4. $\frac{16}{5}$

Question Type : **MCQ**

Question ID : **444792535**

Option 1 ID : **4447921822**

Option 2 ID : **4447921824**

Option 3 ID : **4447921825**

Option 4 ID : **4447921823**

Q.10 Let the lines $L_1 : \vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$, $\lambda \in \mathbb{R}$ and $L_2 : \vec{r} = (4\hat{i} + \hat{j}) + \mu(5\hat{i} + 2\hat{j} + \hat{k})$, $\mu \in \mathbb{R}$, intersect at the point R. Let P and Q be the points lying on lines L_1 and L_2 , respectively, such that $|\overline{PR}| = \sqrt{29}$ and $|\overline{PQ}| = \sqrt{\frac{47}{3}}$. If the point P lies in the first octant, then $27(QR)^2$ is equal to

- Options 1. 340
2. 348
3. 360
4. 320

Question Type : MCQ

Question ID : 444792541

Option 1 ID : 4447921847

Option 2 ID : 4447921848

Option 3 ID : 4447921849

Option 4 ID : 4447921846

Q.11 Let $S = \frac{1}{25!} + \frac{1}{3!23!} + \frac{1}{5!21!} + \dots$ up to 13 terms. If $13S = \frac{2^k}{n!}$, $k \in \mathbb{N}$, then $n + k$ is equal to

- Options 1. 49
2. 51
3. 52
4. 50

Question Type : MCQ

Question ID : 444792532

Option 1 ID : 4447921810

Option 2 ID : 4447921812

Option 3 ID : 4447921813

Option 4 ID : 4447921811

Q.12 If the domain of the function

$$f(x) = \log_{(10x^2 - 17x + 7)}(18x^2 - 11x + 1)$$

is $(-\infty, a) \cup (b, c) \cup (d, \infty) - \{e\}$, then

$90(a + b + c + d + e)$ equals:

- Options 1. 316
2. 170
3. 177
4. 307

Question Type : MCQ

Question ID : 444792527

Option 1 ID : 4447921790

Option 2 ID : 4447921792

Option 3 ID : 4447921793

Option 4 ID : 4447921791

Q.13 Let A_1 be the bounded area enclosed by the curves $y = x^2 + 2$, $x + y = 8$ and y -axis that lies in the first quadrant. Let A_2 be the bounded area enclosed by the curves $y = x^2 + 2$, $y^2 = x$, $x = 2$, and y -axis that lies in the first quadrant. Then $A_1 - A_2$ is equal to

- Options
1. $\frac{2}{3}(3\sqrt{2} + 1)$
 2. $\frac{2}{3}(\sqrt{2} + 1)$
 3. $\frac{2}{3}(4\sqrt{2} + 1)$
 4. $\frac{2}{3}(2\sqrt{2} + 1)$

Question Type : **MCQ**

Question ID : **444792545**

Option 1 ID : **4447921864**

Option 2 ID : **4447921862**

Option 3 ID : **4447921865**

Option 4 ID : **4447921863**

Q.14 Let 729, 81, 9, 1, ... be a sequence and P_n denote the product of the first n terms of this sequence.

If $2 \sum_{n=1}^{40} (P_n)^{\frac{1}{n}} = \frac{3^\alpha - 1}{3^\beta}$ and $\gcd(\alpha, \beta) = 1$, then

$\alpha + \beta$ is equal to

- Options
1. **75**
 2. **73**
 3. **74**
 4. **76**

Question Type : **MCQ**

Question ID : **444792529**

Option 1 ID : **4447921800**

Option 2 ID : **4447921798**

Option 3 ID : **4447921799**

Option 4 ID : **4447921801**

Q.15

$$\text{Let } f(t) = \int \left(\frac{1 - \sin(\log_e t)}{1 - \cos(\log_e t)} \right) dt, \quad t > 1.$$

If $f(e^{\pi/2}) = -e^{\pi/2}$ and $f(e^{\pi/4}) = \alpha e^{\pi/4}$, then α equals

- Options
1. $-1 - 2\sqrt{2}$
 2. $1 + \sqrt{2}$
 3. $-1 - \sqrt{2}$
 4. $-1 + \sqrt{2}$

Question Type : MCQ

Question ID : 444792544

Option 1 ID : 4447921858

Option 2 ID : 4447921860

Option 3 ID : 4447921859

Option 4 ID : 4447921861

Q.16 The number of the real solutions of the equation:

$$x|x+3| + |x-1| - 2 = 0 \text{ is}$$

- Options
1. 2
 2. 5
 3. 4
 4. 3

Question Type : MCQ

Question ID : 444792530

Option 1 ID : 4447921802

Option 2 ID : 4447921804

Option 3 ID : 4447921805

Option 4 ID : 4447921803

Q.17 From a lot containing 10 defective and 90 non-defective bulbs, 8 bulbs are selected one by one with replacement. Then the probability of getting at least 7 defective bulbs is

- Options
1. $\frac{7}{10^7}$
 2. $\frac{81}{10^8}$
 3. $\frac{73}{10^8}$
 4. $\frac{67}{10^8}$

Question Type : MCQ

Question ID : 444792534

Option 1 ID : 4447921819

Option 2 ID : 4447921820

Option 3 ID : 4447921818

Option 4 ID : 4447921821

Q.18 The mean and variance of a data of 10 observations are 10 and 2, respectively. If an observations α in this data is replaced by β , then the mean and variance become 10.1 and 1.99, respectively. Then $\alpha + \beta$ equals

- Options 1. 20
2. 5
3. 10
4. 15

Question Type : MCQ

Question ID : 444792533

Option 1 ID : 4447921817

Option 2 ID : 4447921814

Option 3 ID : 4447921815

Option 4 ID : 4447921816

Q.19 Let $S = \left\{ z \in \mathbb{C} : \left| \frac{z-6i}{z-2i} \right| = 1 \text{ and } \left| \frac{z-8+2i}{z+2i} \right| = \frac{3}{5} \right\}$.

Then $\sum_{z \in S} |z|^2$ is equal to

- Options 1. 423
2. 385
3. 413
4. 398

Question Type : MCQ

Question ID : 444792528

Option 1 ID : 4447921797

Option 2 ID : 4447921794

Option 3 ID : 4447921796

Option 4 ID : 4447921795

Q.20 The value of $\frac{\sqrt{3} \operatorname{cosec} 20^\circ - \sec 20^\circ}{\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ}$ is equal to

- Options 1. 12
2. 64
3. 32
4. 16

Question Type : MCQ

Question ID : 444792539

Option 1 ID : 4447921839

Option 2 ID : 4447921841

Option 3 ID : 4447921840

Option 4 ID : 4447921838

Section : Mathematics Section B

Q.21

Let a differentiable function f satisfy the equation $\int_0^{36} f\left(\frac{tx}{36}\right) dt = 4\alpha f(x)$.

If $y = f(x)$ is a standard parabola passing through the points $(2, 1)$ and $(-4, \beta)$, then β^α is equal to _____.

Question Type : SA

Question ID : 444792550

Q.22 The number of numbers greater than 5000, less than 9000 and divisible by 3, that can be formed using the digits 0, 1, 2, 5, 9, if the repetition of the digits is allowed, is _____

Question Type : SA

Question ID : 444792547

Q.23 The number of 3×2 matrices A, which can be formed using the elements of the set $\{-2, -1, 0, 1, 2\}$ such that the sum of all the diagonal elements of $A^T A$ is 5, is _____

Question Type : SA

Question ID : 444792546

Q.24 Let $(2\alpha, \alpha)$ be the largest interval in which the function $f(t) = \frac{|t+1|}{t^2}$, $t < 0$, is strictly decreasing. Then the local maximum value of the function $g(x) = 2 \log_e (x-2) + \alpha x^2 + 4x - \alpha$, $x > 2$, is _____

Question Type : SA

Question ID : 444792549

Q.25 Let a line L passing through the point P (1,1,1) be perpendicular to the lines $\frac{x-4}{4} = \frac{y-1}{1} = \frac{z-1}{1}$ and $\frac{x-17}{1} = \frac{y-71}{1} = \frac{z}{0}$. Let the line L intersect the yz-plane at the point Q. Another line parallel to L and passing through the point S (1, 0, -1) intersects the yz-plane at the point R. Then the square of the area of the parallelogram PQRS is equal to _____.

Question Type : SA

Question ID : 444792548

Section : Physics Section A

Q.26 Density of water at 4 °C and 20 °C are 1000 kg/m³ and 998 kg/m³ respectively. The increase in internal energy of 4 kg of water when it is heated from 4 °C to 20 °C is _____ J.
(specific heat capacity of water = 4.2 J / kg. and 1 atmospheric pressure = 10⁵ Pa)

- Options 1. 268799.2
2. 234699.2
3. 315826.2
4. 258700.8

Question Type : MCQ

Question ID : 444792557

Option 1 ID : 4447921897

Option 2 ID : 4447921898

Option 3 ID : 4447921896

Option 4 ID : 4447921895

Q.27 Three masses 200 kg, 300 kg and 400 kg are placed at the vertices of an equilateral triangle with sides 20 m. They are rearranged on the vertices of a bigger triangle of side 25 m and with the same centre. The work done in this process _____ J.
(Gravitational constant $G = 6.7 \times 10^{-11} \text{ N m}^2/\text{kg}^2$)

- Options 1. 1.74×10^{-7}
2. 2.85×10^{-7}
3. 9.86×10^{-6}
4. 4.77×10^{-7}

Question Type : **MCQ**

Question ID : **444792554**

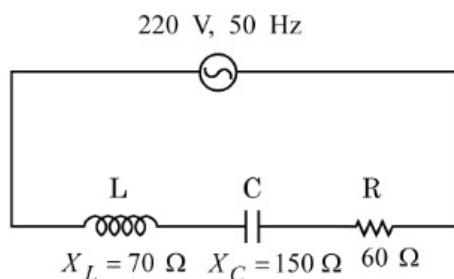
Option 1 ID : **4447921886**

Option 2 ID : **4447921885**

Option 3 ID : **4447921884**

Option 4 ID : **4447921883**

Q.28 For the series *LCR* circuit connected with 220 V, 50 Hz a.c source as shown in the figure, the power factor is $\frac{\alpha}{10}$. The value of α is _____.



- Options 1. 10
2. 8
3. 6
4. 4

Question Type : **MCQ**

Question ID : **444792562**

Option 1 ID : **4447921918**

Option 2 ID : **4447921917**

Option 3 ID : **4447921916**

Option 4 ID : **4447921915**

Q.29 In a microscope of tube length 10 cm two convex lenses are arranged with focal length of 2 cm and 5 cm. Total magnification obtained with this system for normal adjustment is $(5)^k$. The value of k is _____.

- Options 1. 4
2. 3.5
3. 5
4. 2

Question Type : **MCQ**

Question ID : **444792567**

Option 1 ID : **4447921935**

Option 2 ID : **4447921938**

Option 3 ID : **4447921937**

Option 4 ID : **4447921936**

Q.30 A brass wire of length 2 m and radius 1 mm at 27 °C is held taut between two rigid supports. Initially it was cooled to a temperature of –43 °C creating a tension T in the wire. The temperature to which the wire has to be cooled in order to increase the tension in it to $1.4T$, is _____ °C.

- Options
1. –71
 2. –65
 3. –86
 4. –80

Question Type : MCQ

Question ID : 444792558

Option 1 ID : 4447921900

Option 2 ID : 4447921902

Option 3 ID : 4447921899

Option 4 ID : 4447921901

Q.31 Two electrons are moving in orbits of two hydrogen like atoms with speeds 3×10^5 m/s and 2.5×10^5 m/s respectively. If the radii of these orbits are nearly same then the possible order of energy states are _____ respectively.

- Options
1. 8 and 10
 2. 9 and 8
 3. 6 and 5
 4. 10 and 12

Question Type : MCQ

Question ID : 444792570

Option 1 ID : 4447921950

Option 2 ID : 4447921947

Option 3 ID : 4447921949

Option 4 ID : 4447921948

Q.32 A spring of force constant 15 N/m is cut into two pieces. If the ratio of their length is 1:3, then the force constant of smaller piece is _____ N/m.

- Options
1. 20
 2. 60
 3. 45
 4. 15

Question Type : MCQ

Question ID : 444792552

Option 1 ID : 4447921875

Option 2 ID : 4447921877

Option 3 ID : 4447921876

Option 4 ID : 4447921878

Q.33 Two masses 400 g and 350 g are suspended from the ends of a light string passing over a heavy pulley of radius 2 cm. When released from rest the heavier mass is observed to fall 81 cm in 9 s. The rotational inertia of the pulley is _____ kg.m^2 .
($g = 9.8 \text{ m/s}^2$)

- Options 1. 8.3×10^{-3}
2. 1.86×10^{-2}
3. 4.75×10^{-3}
4. 9.5×10^{-3}

Question Type : MCQ

Question ID : 444792556

Option 1 ID : 4447921891

Option 2 ID : 4447921893

Option 3 ID : 4447921894

Option 4 ID : 4447921892

Q.34 Two resistors of 100Ω each are connected in series with a 9 V battery. A voltmeter of 400Ω resistance is connected to measure the voltage drop across one of the resistors. The voltmeter reading is _____ V.

- Options 1. 4
2. 3
3. 4.5
4. 2

Question Type : MCQ

Question ID : 4447921913

Option 1 ID : 4447921913

Option 2 ID : 4447921912

Option 3 ID : 4447921914

Option 4 ID : 4447921911

Q.35 There are three co-centric conducting spherical shells A , B and C of radii a , b and c respectively ($c > b > a$) and they are charged with charge q_1 , q_2 and q_3 respectively. The potentials of the spheres A , B and C respectively, are :

Options 1.

$$\frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2 + q_3}{a} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2 + q_3}{b} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2 + q_3}{c} \right)$$

2.

$$\frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{a} + \frac{q_2}{b} + \frac{q_3}{c} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2}{b} + \frac{q_3}{c} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2 + q_3}{c} \right)$$

3.

$$\frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{a} + \frac{q_2}{b} + \frac{q_3}{c} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2 + q_3}{b} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2 + q_3}{c} \right)$$

4.

$$\frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2 + q_3}{a} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2}{b} + \frac{q_3}{c} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{a} + \frac{q_2}{b} + \frac{q_3}{c} \right)$$

Question Type : MCQ

Question ID : 444792563

Option 1 ID : 4447921921

Option 2 ID : 4447921920

Option 3 ID : 4447921919

Option 4 ID : 4447921922

Q.36 Match the LIST-I with LIST-II

List-I		List-II	
A.	Radio-wave	I.	is produced by Magnetron valve
B.	Micro-wave	II.	due to change in the vibrational modes of atoms
C.	Infrared-wave	III.	due to inner shell electrons moving from higher energy level to lower energy level
D.	X-ray	IV.	due to rapid acceleration of electrons

Choose the *correct* answer from the options given below:

Options 1. A-IV, B-II, C-I, D-III

2. A-IV, B-III, C-I, D-II

3. A-II, B-IV, C-III, D-I

4. A-IV, B-I, C-II, D-III

Question Type : MCQ

Question ID : 444792565

Option 1 ID : 4447921930

Option 2 ID : 4447921927

Option 3 ID : 4447921929

Option 4 ID : 4447921928

Q.37 Three charges $+2q$, $+3q$ and $-4q$ are situated at $(0, -3a)$, $(2a, 0)$ and $(-2a, 0)$ respectively in the $x y$ plane. The resultant dipole moment about origin is _____.

Options 1. $2qa(3\hat{j} - 7\hat{i})$

2. $2qa(7\hat{i} - 3\hat{j})$

3. $2qa(3\hat{i} - 7\hat{j})$

4. $2qa(3\hat{j} - \hat{i})$

Question Type : MCQ

Question ID : 444792564

Option 1 ID : 4447921926

Option 2 ID : 4447921923

Option 3 ID : 4447921925

Option 4 ID : 4447921924

Q.38 Given below are two statements:

Statement I: For all elements, greater the mass of the nucleus, greater is the binding energy per nucleon.

Statement II: For all elements, nuclei with less binding energy per nucleon transforms to nuclei with greater binding energy per nucleon.

In the light of the above statements, choose the *correct* answer from the options given below

- Options
1. Both Statement I and Statement II are true
 2. Statement I is false but Statement II is true
 3. Statement I is true but Statement II is false
 4. Both Statement I and Statement II are false

Question Type : MCQ

Question ID : 444792569

Option 1 ID : 4447921943

Option 2 ID : 4447921946

Option 3 ID : 4447921945

Option 4 ID : 4447921944

Q.39 The electrostatic potential in a charged spherical region of radius r varies as

$V = ar^3 + b$, where a and b are constants. The total charge in the sphere of unit radius is $\alpha \times \pi a \epsilon_0$. The value of α is _____.

(permittivity of vacuum is ϵ_0)

- Options
1. - 9
 2. - 8
 3. - 12
 4. - 6

Question Type : MCQ

Question ID : 444792560

Option 1 ID : 4447921910

Option 2 ID : 4447921909

Option 3 ID : 4447921907

Option 4 ID : 4447921908

Q.40 An unpolarised light is incident at an interface of two dielectric media having refractive indices of 2 (incident medium) and $2\sqrt{3}$ (medium) respectively. To satisfy the condition that reflected and refracted rays are perpendicular to each other, the angle of incidence is _____.

- Options
1. 60°
 2. 10°
 3. 30°
 4. 45°

Question Type : MCQ

Question ID : 444792566

Option 1 ID : 4447921931

Option 2 ID : 4447921934

Option 3 ID : 4447921933

Option 4 ID : 4447921932

Q.41 A cylindrical block of mass M and area of cross section A is floating in a liquid of density ρ and with its axis vertical. When depressed a little and released the block starts oscillating. The period of oscillation is _____.

Options

1. $2\pi\sqrt{\frac{M}{\rho Ag}}$

2. $\pi\sqrt{\frac{2M}{\rho Ag}}$

3. $\pi\sqrt{\frac{\rho A}{Mg}}$

4. $2\pi\sqrt{\frac{\rho A}{Mg}}$

Question Type : **MCQ**

Question ID : **444792559**

Option 1 ID : **4447921903**

Option 2 ID : **4447921904**

Option 3 ID : **4447921906**

Option 4 ID : **4447921905**

Q.42 A boy throws a ball into air at 45° from the horizontal to land it on a roof of a building of height H . If the ball attains maximum height in 2 s and lands on the building in 3 s after launch, then value of H is _____m.

($g = 10 \text{ m/s}^2$)

Options

1. 15

2. 25

3. 20

4. 10

Question Type : **MCQ**

Question ID : **444792555**

Option 1 ID : **4447921887**

Option 2 ID : **4447921888**

Option 3 ID : **4447921889**

Option 4 ID : **4447921890**

Q.43 Match the LIST-I with LIST-II

List-I		List-II	
A.	Magnetic induction	I.	$MLT^{-2}A^{-2}$
B.	Magnetic flux	II.	$ML^2T^{-2}A^{-2}$
C.	Magnetic permeability	III.	$ML^0T^{-2}A^{-1}$
D.	Self inductance	IV.	$ML^2T^{-2}A^{-1}$

Choose the *correct* answer from the options given below:

- Options 1. A-III, B-IV, C-I, D-II
2. A-I, B-III, C-IV, D-II
3. A-IV, B-III, C-I, D-II
4. A-III, B-IV, C-II, D-I

Question Type : MCQ

Question ID : 444792551

Option 1 ID : 4447921874

Option 2 ID : 4447921873

Option 3 ID : 4447921872

Option 4 ID : 4447921871

Q.44 The exit surface of a prism with refractive index n is coated with a material having refractive index $\frac{n}{2}$. When this prism is set for minimum angle of deviation, it exactly meets the condition of critical angle. The prism angle is _____.

- Options 1. 60°
2. 30°
3. 15°
4. 45°

Question Type : MCQ

Question ID : 444792568

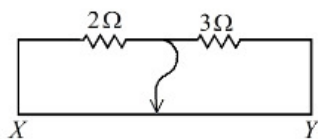
Option 1 ID : 4447921941

Option 2 ID : 4447921940

Option 3 ID : 4447921939

Option 4 ID : 4447921942

- Q.45** Two resistors $2\ \Omega$ and $3\ \Omega$ are connected in the gaps of bridge as shown in figure. The null point is obtained with the contact of jockey at some point on wire XY . When an unknown resistor is connected in parallel with $3\ \Omega$ resistor, the null point is shifted by 22.5 cm toward Y . The resistance of unknown resistor is _____ Ω .



- Options 1. 2
2. 3
3. 1
4. 4

Question Type : MCQ

Question ID : 444792553

Option 1 ID : 4447921880

Option 2 ID : 4447921881

Option 3 ID : 4447921879

Option 4 ID : 4447921882

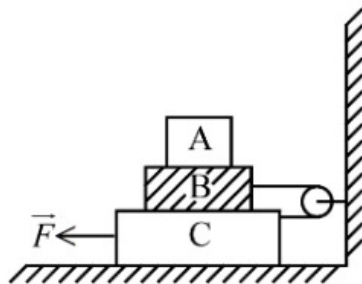
Section : Physics Section B

- Q.46** A gas of certain mass filled in a closed cylinder at a pressure of 3.23 kPa has temperature $50\text{ }^\circ\text{C}$. The gas is now heated to double its temperature. The modified pressure is ____ Pa.

Question Type : SA

Question ID : 444792574

- Q.47** In the given figure the blocks A , B and C weigh 4 kg , 6 kg and 8 kg respectively. The co-efficient of sliding friction between any two surfaces is 0.5 . The force $\vec{F}$ required to slide the block C with constant speed is ____ N. (Use $g = 10\text{ m/s}^2$)



Question Type : SA

Question ID : 444792572

- Q.48** A voltage regulating circuit consisting of Zener diode, having break-down voltage of 10 V and maximum power dissipation of 0.4 W , is operated at 15 V . The approximate value of protective resistance in this circuit is _____ Ω .

Question Type : SA

Question ID : 444792571

- Q.49** Sixty four rain drops of radius 1 mm each falling down with a terminal velocity of 10 cm/s coalesce to form a bigger drop. The terminal velocity of bigger drop is ____ cm/s.

Question Type : SA

Question ID : 444792573

Q.50 A short bar magnet placed with its axis at 30° with an external field of 800 Gauss, experiences a torque of 0.016 N.m. The work done in moving it from most stable to most unstable position is $\alpha \times 10^{-3}$ J. The value of α is _____.

Question Type : SA

Question ID : 444792575

Section : Chemistry Section A

Q.51 Consider three metal chlorides x, y and z, where x is water soluble at room temperature, y is sparingly soluble in water at room temperature and z is soluble in hot water. x, y and z are respectively

- Options 1. AlCl_3 , PbCl_2 and BaCl_2
 2. MgCl_2 , AgCl and AlCl_3
 3. CuCl_2 , AgCl and PbCl_2
 4. AgCl , Hg_2Cl_2 and PbCl_2

Question Type : MCQ

Question ID : 444792595

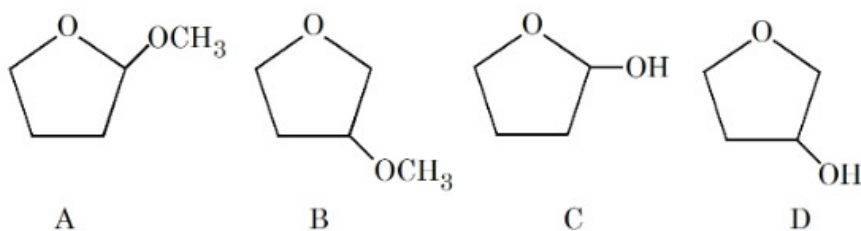
Option 1 ID : 4447922032

Option 2 ID : 4447922035

Option 3 ID : 4447922033

Option 4 ID : 4447922034

Q.52 A student is given one compound among the following compounds that gives positive test with Tollen's reagent.



The compound is :

- Options 1. A
 2. D
 3. B
 4. C

Question Type : MCQ

Question ID : 444792587

Option 1 ID : 4447922000

Option 2 ID : 4447922003

Option 3 ID : 4447922001

Option 4 ID : 4447922002

Q.53 Given below are two statements:

Statement I: The number of paramagnetic species among $[\text{CoF}_6]^{3-}$, $[\text{TiF}_6]^{3-}$, V_2O_5 and $[\text{Fe}(\text{CN})_6]^{3-}$ is 3.

Statement II:

$\text{K}_4[\text{Fe}(\text{CN})_6] < \text{K}_3[\text{Fe}(\text{CN})_6] < [\text{Fe}(\text{H}_2\text{O})_6] \text{SO}_4 \cdot \text{H}_2\text{O} < [\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_3$ is the correct order in terms of number of unpaired electron(s) present in the complexes.

In the light of the above statements, choose the *correct* answer from the options given below

- Options
1. Statement I is true but Statement II is false
 2. Both Statement I and Statement II are false
 3. Statement I is false but Statement II is true
 4. Both Statement I and Statement II are true

Question Type : MCQ

Question ID : 444792586

Option 1 ID : 4447921998

Option 2 ID : 4447921997

Option 3 ID : 4447921999

Option 4 ID : 4447921996

Q.54 A solution is prepared by dissolving 0.3 g of a non-volatile non-electrolyte solute 'A' of molar mass 60 g mol^{-1} and 0.9 g of a non-volatile non-electrolyte solute 'B' of molar mass 180 g mol^{-1} in 100 mL H_2O at 27°C . Osmotic pressure of the solution will be

[Given: $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$]

- Options
1. 1.47 atm
 2. 0.82 atm
 3. 2.46 atm
 4. 1.23 atm

Question Type : MCQ

Question ID : 444792579

Option 1 ID : 4447921970

Option 2 ID : 4447921968

Option 3 ID : 4447921971

Option 4 ID : 4447921969

Q.55 Given below are statements about some molecules/ions.

Identify the **CORRECT** statements.

- A. The dipole moment value of NF_3 is higher than that of NH_3 .
- B. The dipole moment value of BeH_2 is zero.
- C. The bond order of O_2^{2-} and F_2 is same.
- D. The formal charge on the central oxygen atom of ozone is -1 .
- E. In NO_2 , all the three atoms satisfy the octet rule, hence it is very stable.

Choose the **correct** answer from the options given below:

Options 1. B & C Only

- 2. A, C & D Only
- 3. A, B, C, D & E
- 4. B, C & D Only

Question Type : MCQ

Question ID : 444792578

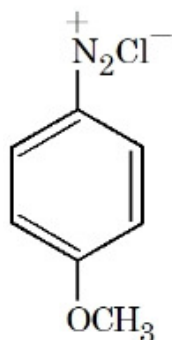
Option 1 ID : 4447921965

Option 2 ID : 4447921967

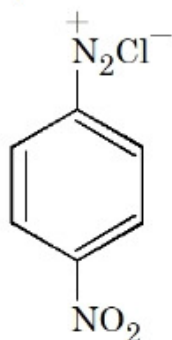
Option 3 ID : 4447921964

Option 4 ID : 4447921966

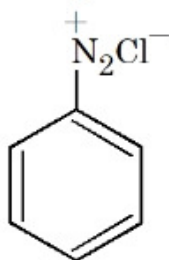
Q.56 The correct stability order of the following diazonium salts is



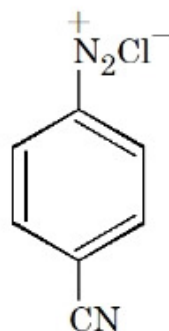
(A)



(B)



(C)



(D)

Options 1. $A > C > D > B$

- 2. $C > D > B > A$
- 3. $A > B > C > D$
- 4. $C > A > D > B$

Question Type : MCQ

Question ID : 444792593

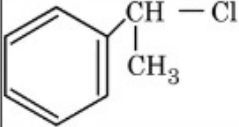
Option 1 ID : 4447922025

Option 2 ID : 4447922027

Option 3 ID : 4447922024

Option 4 ID : 4447922026

Q.57 Match the LIST-I with LIST-II

List-I Chloro derivative		List-II Example	
A.	Vinyl Chloride	I.	$\text{CH}_2 = \text{CH} - \text{CH}_2\text{Cl}$
B.	Benzyl Chloride	II.	$\text{CH}_3 - \text{CH}(\text{Cl})\text{CH}_3$
C.	Alkyl Chloride	III.	$\text{CH}_2 = \text{CHCl}$
D.	Allyl Chloride	IV.	

Choose the **correct** answer from the options given below:

Options 1. A-III, B-IV, C-II, D-I

2. A-I, B-II, C-IV, D-III

3. A-IV, B-I, C-III, D-II

4. A-III, B-IV, C-I, D-II

Question Type : MCQ

Question ID : 444792591

Option 1 ID : 4447922017

Option 2 ID : 4447922016

Option 3 ID : 4447922019

Option 4 ID : 4447922018

Q.58 Given below are two statements:

Statement I: $\text{K} > \text{Mg} > \text{Al} > \text{B}$ is the correct order in terms of metallic character.

Statement II: Atomic radius is always greater than the ionic radius for any element.

In the light of the above statements, choose the **correct** answer from the options given below

Options 1. Both Statement I and Statement II are false

2. Statement I is true but Statement II is false

3. Statement I is false but Statement II is true

4. Both Statement I and Statement II are true

Question Type : MCQ

Question ID : 444792583

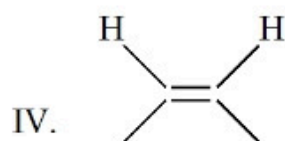
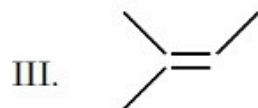
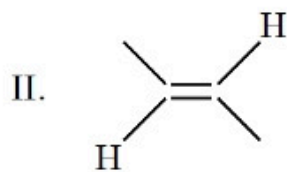
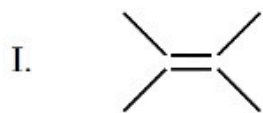
Option 1 ID : 4447921985

Option 2 ID : 4447921986

Option 3 ID : 4447921987

Option 4 ID : 4447921984

Q.59 Arrange the following alkenes in decreasing order of stability.



Choose the **correct** answer from the options given below:

Options 1. III > I > II > IV

2. III > II > I > IV

3. I > III > II > IV

4. I > III > IV > II

Question Type : MCQ

Question ID : 444792589

Option 1 ID : 4447922009

Option 2 ID : 4447922008

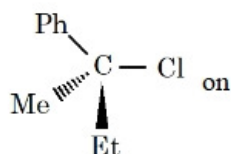
Option 3 ID : 4447922010

Option 4 ID : 4447922011

Q.60 Given below are two statements:

Statement I: 'C - Cl' bond is stronger in $\text{CH}_2 = \text{CH} - \text{Cl}$ than $\text{CH}_3 - \text{CH}_2 - \text{Cl}$

Statement II: The given optically active molecule,



hydrolysis gives a solution that can rotate the plane polarized light.

In the light of the above statements, choose the *correct* answer from the options given below

- Options
1. Statement I is true but Statement II is false
 2. Both Statement I and Statement II are true
 3. Statement I is false but Statement II is true
 4. Both Statement I and Statement II are false

Question Type : MCQ

Question ID : 444792590

Option 1 ID : 4447922014

Option 2 ID : 4447922012

Option 3 ID : 4447922015

Option 4 ID : 4447922013

Q.61 Match the LIST-I with LIST-II

List-I Isothermal process for ideal gas system		List-II Work done ($V_f > V_i$)	
A.	Reversible expansion	I.	$w = 0$
B.	Free expansion	II.	$w = -nRT \ln \frac{V_f}{V_i}$
C.	Irreversible expansion	III.	$w = -p_{\text{ex}}(V_f - V_i)$
D.	Irreversible compression	IV.	$w = -p_{\text{ex}}(V_i - V_f)$

Choose the *correct* answer from the options given below:

- Options
1. A-IV, B-I, C-III, D-II
 2. A-IV, B-II, C-III, D-I
 3. A-II, B-I, C-III, D-IV
 4. A-I, B-III, C-II, D-IV

Question Type : MCQ

Question ID : 444792577

Option 1 ID : 4447921962

Option 2 ID : 4447921960

Option 3 ID : 4447921961

Option 4 ID : 4447921963

Q.62 'W' g of a non-volatile electrolyte solid solute of molar mass 'M' g mol⁻¹ when dissolved in 100 mL water, decreases vapour pressure of water from 640 mm Hg to 600 mm Hg. If aqueous solution of the electrolyte boils at 375 K and K_b for water is 0.52 K kg mol⁻¹, then the mole fraction of the electrolyte solute (x₂) in the solution can be expressed as
(Given : density of water = 1 g/mL and boiling point of water = 373 K)

- Options
1. $\frac{1.3}{8} \times \frac{W}{M}$
 2. $\frac{16}{2.6} \times \frac{W}{M}$
 3. $\frac{2.6}{16} \times \frac{M}{W}$
 4. $\frac{1.3}{8} \times \frac{M}{W}$

Question Type : MCQ

Question ID : 444792580

Option 1 ID : 4447921974

Option 2 ID : 4447921975

Option 3 ID : 4447921973

Option 4 ID : 4447921972

Q.63 Among the following, the CORRECT combinations are

- A. IF₃ → T-shaped (sp³d)
- B. IF₅ → Square pyramidal (sp³d²)
- C. IF₇ → Pentagonal bipyramidal (sp³d³)
- D. ClO₄⁻ → Square planar (sp²d)

Choose the *correct* answer from the options given below:

- Options
1. A and B Only
 2. B, C and D Only
 3. A, B, C and D
 4. A, B and C Only

Question Type : MCQ

Question ID : 444792584

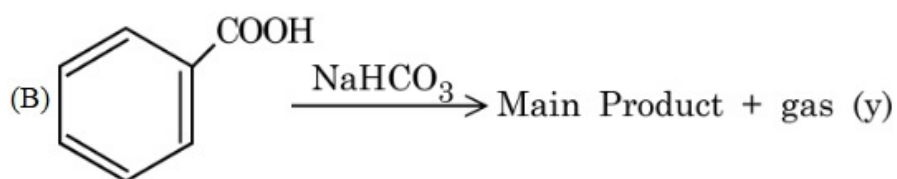
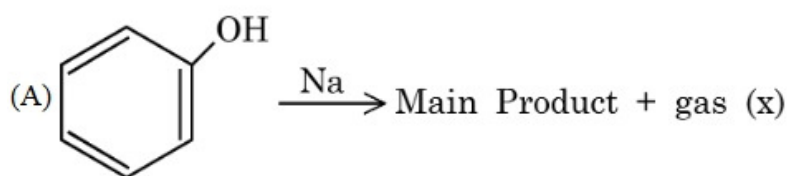
Option 1 ID : 4447921990

Option 2 ID : 4447921989

Option 3 ID : 4447921988

Option 4 ID : 4447921991

Q.64 Consider the following two reactions A and B.



Numerical value of [molar mass of x + molar mass of y] is ____.

- Options
1. 46
 2. 4
 3. 88
 4. 160

Question Type : MCQ

Question ID : 444792592

Option 1 ID : 4447922020

Option 2 ID : 4447922023

Option 3 ID : 4447922022

Option 4 ID : 4447922021

Q.65 Consider a mixture 'X' which is made by dissolving 0.4 mol of $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and 0.4 mol of $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ in water to make 4 L of solution. When 2 L of mixture 'X' is allowed to react with excess of AgNO_3 , it forms precipitate 'Y'. The rest 2 L of mixture 'X' reacts with excess BaCl_2 to form precipitate 'Z'. Which of the following statements is **CORRECT**?

- Options
1. 0.2 mol of 'Z' is formed.
 2. 0.1 mol of 'Y' is formed.
 3. 'Y' is BaSO_4 and 'Z' is AgBr .
 4. 0.4 mol of 'Z' is formed.

Question Type : MCQ

Question ID : 444792576

Option 1 ID : 4447921958

Option 2 ID : 4447921957

Option 3 ID : 4447921959

Option 4 ID : 4447921956

Q.66 Given below are two statements:

Statement I: Hybridisation, shape and spin only magnetic moment of $K_3[Co(CO_3)_3]$ is sp^3d^2 , octahedral and 4.9 BM respectively.

Statement II: Geometry, hybridisation and spin only magnetic moment values (BM) of the ions $[Ni(CN)_4]^{2-}$, $[MnBr_4]^{2-}$ and $[CoF_6]^{3-}$ respectively are square planar, tetrahedral, octahedral; dsp^2 , sp^3 , sp^3d^2 and 0, 5.9, 4.9.

In the light of the above statements, choose the **correct** answer from the options given below

- Options
1. Statement I is true but Statement II is false
 2. Both Statement I and Statement II are false
 3. Both Statement I and Statement II are true
 4. Statement I is false but Statement II is true

Question Type : MCQ

Question ID : 444792585

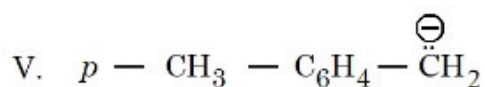
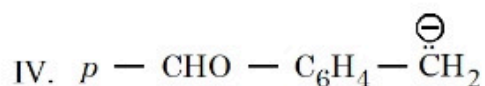
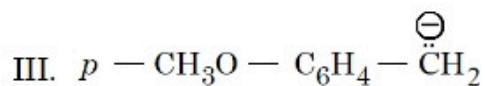
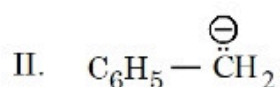
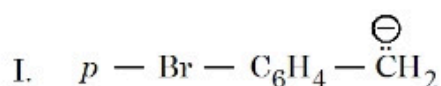
Option 1 ID : 4447921994

Option 2 ID : 4447921993

Option 3 ID : 4447921992

Option 4 ID : 4447921995

Q.67 Arrange the following carbanions in the decreasing order of stability.



Choose the **correct** answer from the options given below:

- Options
1. $I > II > IV > V > III$
 2. $IV > I > II > V > III$
 3. $IV > II > I > III > V$
 4. $I > IV > II > V > III$

Question Type : MCQ

Question ID : 444792588

Option 1 ID : 4447922007

Option 2 ID : 4447922004

Option 3 ID : 4447922005

Option 4 ID : 4447922006

Q.68 $A \rightarrow D$ is an endothermic reaction occurring in three steps (elementary).

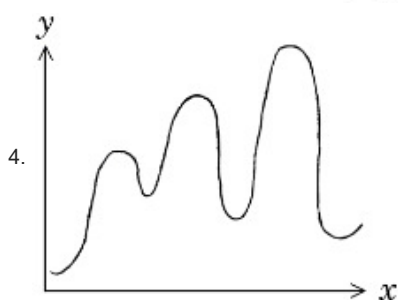
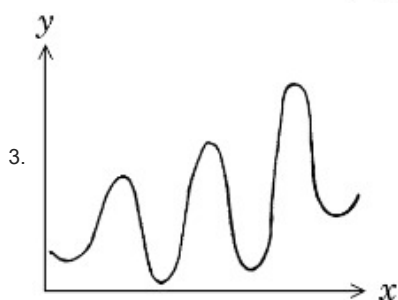
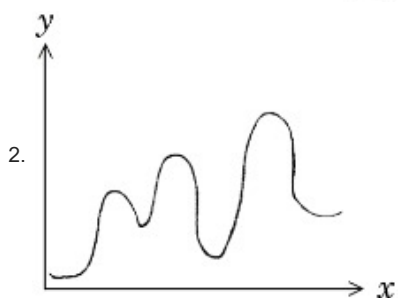
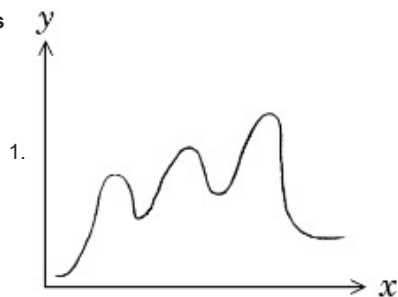
(i) $A \rightarrow B$ $\Delta H_i = +ve$

(ii) $B \rightarrow C$ $\Delta H_{ii} = -ve$

(iii) $C \rightarrow D$ $\Delta H_{iii} = -ve$

Which of the following graphs between potential energy (y-axis) vs reaction coordinate (x-axis) correctly represents the reaction profile of $A \rightarrow D$?

Options



Question Type : MCQ

Question ID : 444792582

Option 1 ID : 4447921983

Option 2 ID : 4447921980

Option 3 ID : 4447921982

Option 4 ID : 4447921981

Q.69 At 27 °C in presence of a catalyst, activation energy of a reaction is lowered by 10 kJ mol⁻¹. The logarithm of ratio of $\frac{k(\text{catalysed})}{k(\text{uncatalysed})}$ is....

(Consider that the frequency factor for both the reactions is same)

- Options 1. 17.41
2. 3.482
3. 0.1741
4. 1.741

Question Type : MCQ

Question ID : 444792581

Option 1 ID : 4447921978

Option 2 ID : 4447921979

Option 3 ID : 4447921976

Option 4 ID : 4447921977

Q.70 A hydroxy compound (X) with molar mass 122 g mol⁻¹ is acetylated with acetic anhydride, using a large excess of the reagent ensuring complete acetylation of all hydroxyl groups. The product obtained has a molar mass of 290 g mol⁻¹. The number of hydroxyl groups present in compound (X) is:

- Options 1. 3
2. 4
3. 2
4. 5

Question Type : MCQ

Question ID : 444792594

Option 1 ID : 4447922028

Option 2 ID : 4447922031

Option 3 ID : 4447922030

Option 4 ID : 4447922029

Section : Chemistry Section B

Q.71 The hydrogen spectrum consists of several spectral lines in Lyman series (L₁, L₂, L₃...; L₁ has lowest energy among Lyman series). Similarly it consists of several spectral lines in Balmer series (B₁, B₂, B₃ ...; B₁ has lowest energy among Balmer lines). The energy of L₁ is x times the energy of B₁. The value of x is _____ × 10⁻¹. (Nearest integer)

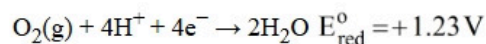
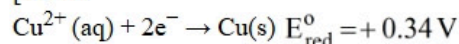
Question Type : SA

Question ID : 444792598

Q.72

Electricity is passed through an acidic solution of Cu^{2+} till all the Cu^{2+} was exhausted, leading to the deposition of 300 mg of Cu metal. However, a current of 600 mA was continued to pass through the same solution for another 28 minutes by keeping the total volume of the solution fixed at 200 mL. The total volume of oxygen evolved at STP during the entire process is _____ mL. (Nearest integer)

[Given:



Molar mass of Cu = 63.54 g mol⁻¹

Molar mass of O₂ = 32 g mol⁻¹

Faraday Constant = 96500 C mol⁻¹

Molar volume at STP = 22.4 L]

Question Type : SA

Question ID : 444792600

Q.73 In Dumas method for estimation of nitrogen, 0.50 g of an organic compound gave 70 mL of nitrogen collected at 300 K and 715 mm pressure. The percentage of nitrogen in the organic compound is ____ %.
(Aqueous tension at 300 K is 15 mm).

Question Type : SA

Question ID : 444792597

Q.74 X and Y are the number of electrons involved, respectively during the oxidation of I^{-} to I_2 and S^{2-} to S by acidified $\text{K}_2\text{Cr}_2\text{O}_7$. The value of X + Y is ____.

Question Type : SA

Question ID : 444792596

Q.75 Consider two Group IV metal ions X^{2+} and Y^{2+} .

A solution containing 0.01 M X^{2+} and 0.01 M Y^{2+} is saturated with H_2S . The pH at which the metal sulphide YS will form as a precipitate is _____. (Nearest integer)

(Given: $K_{\text{sp}}(\text{XS}) = 1 \times 10^{-22}$ at 25°C, $K_{\text{sp}}(\text{YS}) = 4 \times 10^{-16}$ at 25°C,
[H_2S] = 0.1M in solution, $K_{a1} \times K_{a2}(\text{H}_2\text{S}) = 1.0 \times 10^{-21}$, $\log 2 = 0.30$,
 $\log 3 = 0.48$, $\log 5 = 0.70$)

Question Type : SA

Question ID : 444792599